

# BRYAN HEDDLE

Hamilton, Ontario

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## Education

University of Western Ontario

Sep. 2022 – May 2028 | GPA: 3.7

*Mechatronics with AI Systems Engineering*

*London, Ontario*

- **Relevant Coursework:** Data Structures & Algorithms, Statistics, Machine Learning, Deep Learning

## Technical Skills

**Languages:** Python, C, C++, Java, SQL, Bash, JavaScript, HTML/CSS

**ML & CV:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, OpenCV, pandas, NumPy

**Robotics, Embedded & Systems:** ROS 2, NVIDIA Isaac Sim/Lab, CAN, Docker, AWS, Linux, Git, GitLab CI

## Work Experience

Computer Vision Researcher | *VIP Lab, University of Waterloo*

May 2026 – August 2026

- Cut median reprojection error **~92% (15.1 px → 1.2 px)** vs DJI vendor poses by matching RGB drone imagery to 3D LiDAR with SIFT, rejecting outliers with RANSAC, then solving PnP for metric camera poses.
- Lowered building point-to-mesh error from DJI Terra **4.7 cm** to **0.8 cm** with a custom Poisson meshing pipeline.
- Used DN-Splatter Gaussian splatting on those poses to keep views metrically accurate and feed high-quality renders into the change-over-time model, increasing detection confidence.

Embedded Developer Intern | *Geotab*

May 2025 – May 2026

- Built a Python CI job that parses test artifacts and posts Google Chat cards (failed jobs, error links, metadata), then attached an MCP server that pulls extra context per failing test and writes an AI-generated root-cause summary.
- Freed **24+** USB ports on the test server by moving Nordic nRF dongles onto the rigs over USB/IP with a test-framework interface, patching and compiling the Linux vhci driver to raise port limits.
- Expanded hardware test coverage to **4 new products** by building and validating **10+** Raspberry Pi test-farm rigs, installing custom PCBs and wiring, deploying images with Ansible, and configuring each device before rack install.
- Cut serial-related test failures **nearly 80%** by designing a serial interface in the test framework that communicates with a gRPC service, routing traffic from the server to each test rig.

Machine Learning Engineer Intern | *Outlier AI*

May 2024 – August 2024

- Improved response consistency with RLHF by ranking preferred LLM answers from reasoning traces and prompt logs.
- Built RAG evaluation pipelines with unit and regression tests to catch retrieval and answer-quality issues.

## Design Teams

President & Lead Software Engineer

September 2023 – Present

*Western Engineering Mars Rover Team*

*London, Ontario*

- Cut rover teleop video latency from **240 ms** to **44 ms** vs the previous Python stack with a C++ libdatachannel pipeline on the Jetson Orin Nano to a React frontend.
- Implemented CAN for multi-actuator messaging and exposed devices to ROS 2 through a hardware abstraction layer.
- Architected a C++ ros2\_control IK plugin for rover swivels, optimizing continuous rotation and shortest angular path.

## Research & Projects

ServoSim | **PI: Dr. Elvis Chen**

- Authored and won an NVIDIA Academic Grant proposal for sim-to-real robotics research, securing **~\$15K** in hardware (RTX PRO 6000 Max-Q, Jetson AGX Orin).
- Assembled a weighted lever-arm rig to measure torque and gearbox backlash on STS3235 and STS3250 servos, releasing the characterization dataset used to train the actuator models.
- Building ServoSim, an open-source Isaac Lab extension with learned actuator nets and hardware-calibrated domain randomization for position-controlled serial-bus servos.

ERGOS: Dynamic Humanoid Robot | [GitHub](#)

- Raised **\$5K+** in sponsorships by pitching ERGOS as a long-term humanoid robotics research platform.
- Designed and built a **20-DOF** humanoid from scratch, including mechanical design, a custom PCB, embedded electronics, and control.
- Integrated NVIDIA Isaac Sim and ROS 2 as the sim-to-real training and control stack for locomotion.